

# Machine Learning-Based Network Intrusion Detection System

Summer Bachelor Semester Project

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## ABSTRACT

The project is developed by student **João Bernardo Sousa Faria**, under the supervision of **Salima Lamsiyah**.

The project aims to analyze network traffic data and detect cyberattacks using machine learning classification techniques.

To achieve this objective, the system processes labeled network traffic records from the CICIDS2017 dataset, which contains both benign and malicious traffic samples covering attack types such as DDoS, brute force, and port scanning. The study compares the performance of several supervised learning models, such as Random Forest and other classical machine learning classifiers, with the possibility of testing a simple neural network-based approach. These models are evaluated against a rule-based baseline to assess their ability to distinguish malicious traffic from normal traffic.

To assess model performance and interpret results, metrics such as precision, recall, F1-score and confusion matrices are used. In addition, feature importance analysis is applied to identify which network traffic characteristics are most indicative of attacks, and dimensionality reduction techniques such as PCA may be used to visualize the separation between traffic classes.

Overall, the project evaluates the effectiveness of machine learning in detecting network intrusions and examines the trade-offs between detection accuracy and false positive rates in a realistic threat detection scenario.

## Overview of the Scientific Section

The project explores the research question: *"Can machine learning models effectively detect and classify network intrusion attempts from traffic data, and how do they compare to rule-based detection baselines?"*

The goal is to understand how different attack types manifest as statistical patterns in network traffic features, and whether supervised learning can reliably identify them.

The study investigates which features are most discriminative across attack categories, and how model performance varies depending on attack type and class imbalance in the dataset.

## Overview of the Technical Section

The technical section focuses on building and evaluating a machine learning pipeline for network intrusion detection. The planned tools and techniques are as follows:

- **Main Programming Language:** Python;
- **Dataset:** CICIDS2017, containing labeled network traffic flows covering benign traffic and multiple attack categories;
- **Data Processing and Preparation:** libraries such as NumPy and Pandas;
- **Classification Models:** scikit-learn models such as Random Forest and other supervised classifiers, with the possibility of testing a simple neural network model;
- **Evaluation Metrics:** metrics such as precision, recall, F1-score, confusion matrix, and ROC curves;
- **Feature Analysis:** feature importance ranking techniques to identify key indicators of malicious traffic;
- **Dimensionality Reduction:** PCA for visualizing traffic class separability;
- **Visualization:** libraries such as Matplotlib and/or Seaborn for plotting results.